

FCC 3101 CHAPTER 2

HISTORY OF CIVIL ENGINEERING

PART 1

2.1 HISTORICAL BACKGROUND

- 2.1.1 “Civil engineering works” can be traced back to the beginning of **human civilization**.
- 2.1.2 Before 1750, construction was done by **artisans (masons and carpenters)** who would become “**master builders**” or “**architect-engineers**”.
- 2.1.3 **Civil engineering** separated from **architecture** and became a distinct profession in **1750**.
- 2.1.4 Before 1750, knowledge and skills were acquired through experience and apprenticeship, but there was limited scientific input or artistic creativity.

- 2.1.5 Building designs were repetitive, based on what was known to work and with limited increase in scale.
- 2.1.6 Since 1750, **architects** deal with the **artistic** aspects, while **civil engineers** deal with the **technical** aspects of buildings.
- 2.1.7 Civil engineers apply scientific principles and use mathematics to analyse and design structures. This has resulted in safer and bigger designs.
- 2.1.8 This chapter outlines the history of civil engineering from the point of view of construction of buildings from **pre-historic time** to the **current status**.

2.1.9 The chapter is divided into seven historical periods:

1. Stone age (10,000 to 5,000 BC) **aka Neolithic age**
2. Bronze age (5,000 to 1,000 BC)
3. Iron age (1,000 to 50 BC)
4. Middle ages (50 BC to 1,400 AD)
5. Renaissance (1,400 to 1,600 AD)
6. Modern ages (1,600 to 2,000 AD)
7. Current status (2000 to 2023)

2.1.10 In each period four parameters are discussed: -

1. The available **construction materials**
2. The **tools** and **technology** used
3. The level of **scientific** and **mathematical** input
4. The scope of **construction works**

2.2 STONE AGE

- 2.2.1 The construction materials were **stones, adobe, bones, hide and tree logs**.
- 2.2.2 The tools used were made of stone or bone and included **axe, hammer, sickle and chisel**.
- 2.2.3 Construction was done by local inhabitants using traditional “**cave-like**” **designs** and construction methods referred to as “**vernacular architecture**”.
- 2.2.4 The construction works were temporary **grass thatched mud huts and tree log bridges**.

Stone Age Gallery



Figure 2.1 Stone Axe with a wooden handle



Figure 2.2 Hammer made from bone and wood



Figure 2.3 Sickle for cutting thatching grass



Figure 2.4 Chisel made from bone

2.3 BRONZE AGE

- 2.3.1 The mining and refining of metals (copper, tin and zinc) marked the beginning of “true civilization”.
- 2.3.2 **Bronze** was made by alloying **copper** with **tin**.
(**Brass** was made by alloying **copper** with **zinc**).
- 2.3.3 **Bronze** was found to be more durable and to make better tools. The tools included **axe**, **hammer**, **sickle** and **chisel**. A new tool: the **saw** was invented.
- 2.3.4 Construction materials included **stones**, **lime mortar**, **brick** and **timber**.
- 2.3.5 **Written language** started to develop.

- 2.3.6 Five simple machines (**lever, wheel, pulley, inclined plane** and **screw**) were invented.
- 2.3.7 Artisanshship (**masonry, carpentry** and **metallurgy**) started to develop.
- 2.3.8 Construction works included **houses, cities, temples, pavements, bridges** and **water canals**.
- 2.3.9 **Imhotep** (2700 – 2600 BC), who lived in the bronze age, is considered to be **the first architect-engineer known by name**. [Imhotep is believed to have supervised the construction of the **step pyramids** for pharaoh **Djoser** in Egypt.

2.4 IRON AGE

- 2.4.1 **Iron and steel** (an alloy of **iron** with **carbon**) replaced copper and bronze in making tools.
- 2.4.2 Construction materials were **stones**, **lime mortar**, **burnt clay bricks** and **tiles**, **timber** and **glass**.
- 2.4.3 **Lead** was used to make water supply and waste water **pipes**.
- 2.4.4 Three new tools were invented: the **hand plane** for wood working, the **crane** for hoisting heavy loads and the **plumb bob** for maintaining verticality.

- 2.4.5 The **oven** for burning clay bricks and the **lathe** for shaping metal items were also invented.
- 2.4.6 Artisanship (**masonry**, **carpentry** and **metallurgy**) continued to develop.
- 2.4.7 Construction works included houses, cities, temples, palaces, silos, pavements, bridges, canals, aqueducts and drainage for flood water and human waste.
- 2.4.8 **Science** and **mathematics** emerged as part of philosophy. Notable contributors include: -

2.4.9 Pythagoras (570 – 495 BC) A Greek philosopher and mathematician who is credited with deriving the relationship between the three sides of a right angled triangle, referred to as **Pythagoras theorem**.

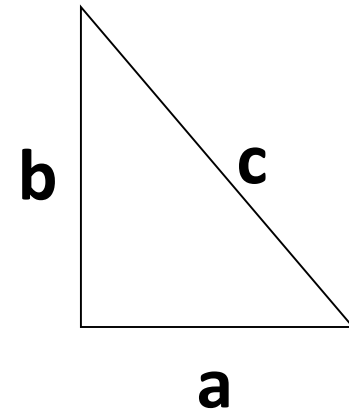


Figure 2.5

Pythagoras Theorem

$$a^2 + b^2 = c^2$$

2.4.10 **Euclid** (323 – 283 BC)

A Greek philosopher and mathematicians who is recognized as the founder of **geometry**.

Euclid published pioneering work in spherical geometry, conic sections, number theory and mathematical rigor.

2.4.11 **Archimedes** (287 – 212 BC)

A Greek philosopher and mathematician who is recognized as one of the pioneer scientists in history.

Archimedes contributions include the **law of the lever** and **Archimedes principle** in hydrostatics.

2.5 MIDDLE AGES

- 2.5.1 There are no recorded changes in construction materials. Stones, hydraulic lime mortar, burnt clay bricks, burnt clay roofing tiles, timber, glass, iron, steel and lead continued to be used.
- 2.5.2 There were no new tools or new technologies invented.
- 2.5.3 Construction works included dwellings, cities, fortifications, castles, cathedrals, palaces, pavements, bridges, water canals, silos, aqueducts and drainage for flood water and human waste.

- 2.5.4 There was no much advancement in science and mathematics. The only notable contribution came from **Vitruvius** (75 BC – 15 AD): A Roman architect-engineer who pioneered **Roman architecture**.
- 2.5.5 **Vitruvius** is credited with originating the idea that all buildings should have three attributes: **Strength, functionality and aesthetics**.

2.6 RENAISSANCE

- 2.6.1 There are no recorded changes in construction materials. Stones, hydraulic lime mortar, burnt clay bricks, burnt clay roofing tiles, timber, glass, iron, steel and lead continued to be used.
- 2.6.2 There were no new tools or new technologies.
- 2.6.3 Construction works included dwellings, cities, fortifications, castles, cathedrals, palaces, pavements, bridges, water canals, silos, aqueducts and drainage for flood water and human waste.

2.6.4 Architecture developed into a profession combining both architecture and engineering. The architect-engineer became the master builder, replacing the masons and carpenters.



Figure 2.6 Florence Cathedral, Italy 1436

2.6.5 There was no much advancement in science and mathematics. The only notable contribution came from **Leonardo da Vinci** (1452 – 1519 AD): An Italian polymath who made significant contributions in architecture and engineering.

2.6.6 **Leonardo da Vinci** designed and produced drawings for the first modern bridge over the Golden Horn in present day Turkey in 1502 AD.

FCC 3101 CHAPTER 2

END OF PART 1